

Energy and Thermodynamics: A Concise Introduction

Copyright 2023 Sample Educational Press. All rights reserved. ISBN 978-0-00-000001-7.

Chapter 1: Energy and Heat

Energy is defined as the capacity of a system to do work or produce heat. Heat is the transfer of thermal energy between systems due to a temperature difference.

The principle of conservation of energy states that the total energy of an isolated system remains constant. Work is the transfer of energy by a force acting through a displacement, given by $W = F d$.

For example, a heated gas expands and does work on its surroundings.

Chapter 2: Laws of Thermodynamics

The first law of thermodynamics states that the change in internal energy of a system equals the heat added minus the work done by the system. This principle is a statement of energy conservation.

Entropy is defined as a measure of the disorder of a system. The second law of thermodynamics states that the entropy of an isolated system never decreases.

For example, heat flows spontaneously from a hot body to a cold body, never the reverse.